Extremes for metocean: Multivariate

Philip Jonathan

Lancaster University

IOGP, London, September 2026
(slides at *www.lancs.ac.uk/∼jonathan)*



Overview of masterclass

- A: Fundamentals
- B: Covariates
- C: Multivariate
- D: covXtreme
- E: Recent developments & take-aways

Overview of C: Multivariate

- Motivating examples
- Understanding extremal dependence
- Conditional extremes
- What can go wrong?
- Further reading

All models are wrong, some models are useful

Motivating examples

- "Associated values" e.g. for $U_{10}|\{H_S^{sp} = 100\text{-year level}\}$
- "Weibull-log normal model" for the joint distribution (H_S, T_P)
- Copula models
- Temporal evolution of sea state H_S during a storm
- Spatial dependence of extreme H_S^{sp} within a hurricane or typhoon

- All these problems address the extremal dependence between two or more variables

Bivariate extremes

Basics

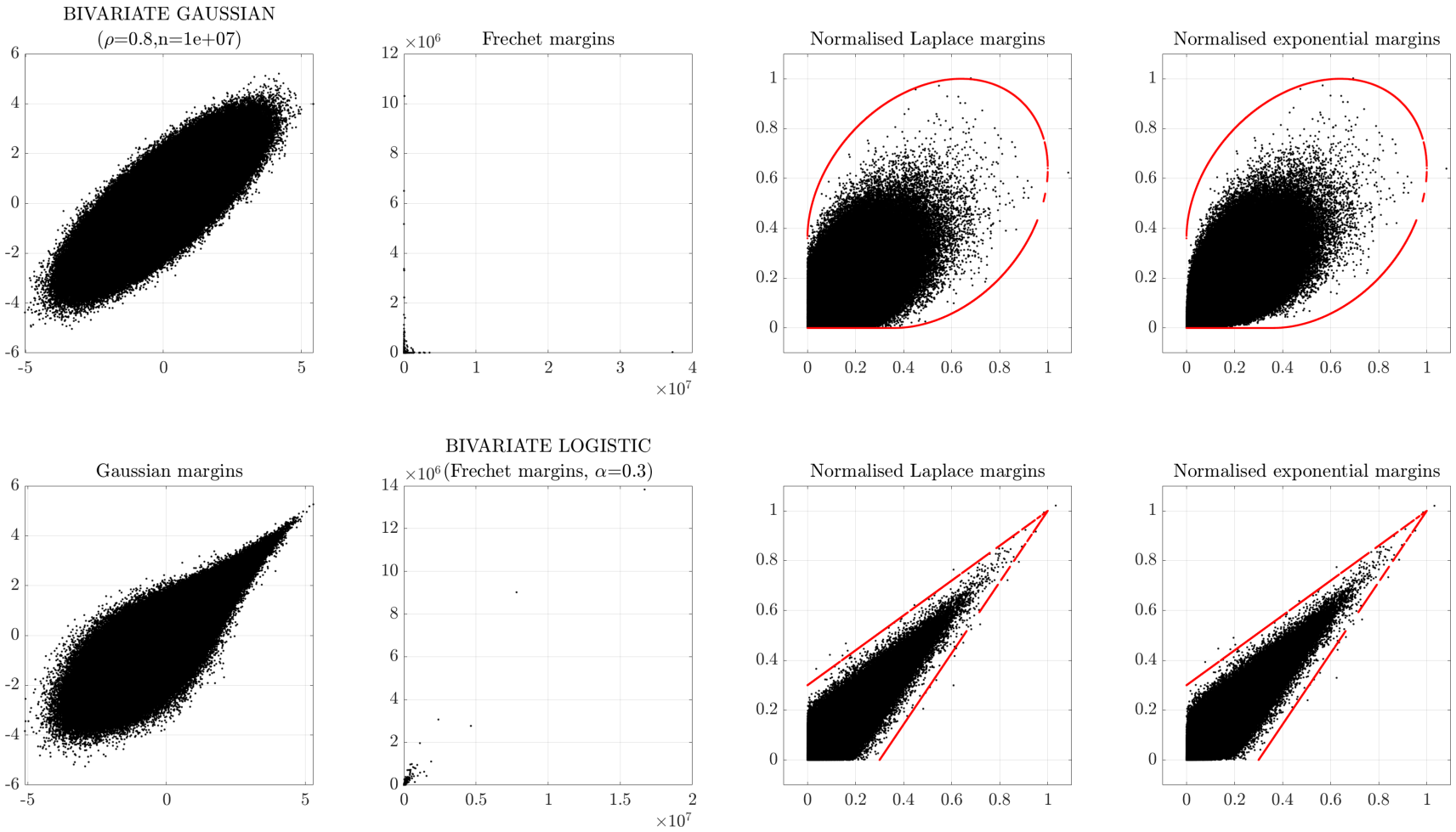
$$\begin{aligned} \text{CDF: } F(x,y) &= C(u,v) \quad \text{where } u = F_X(x) \text{ and } v = F_Y(y) \\ \text{Density: } f_{X,Y}(x,y) &= f_X(x) f_Y(y) \times c(u,v) \end{aligned}$$

- C and c are copula CDF and density, f_X and f_Y are marginal densities
- We are interested in what happens to C and c when u and v are large

Classic examples

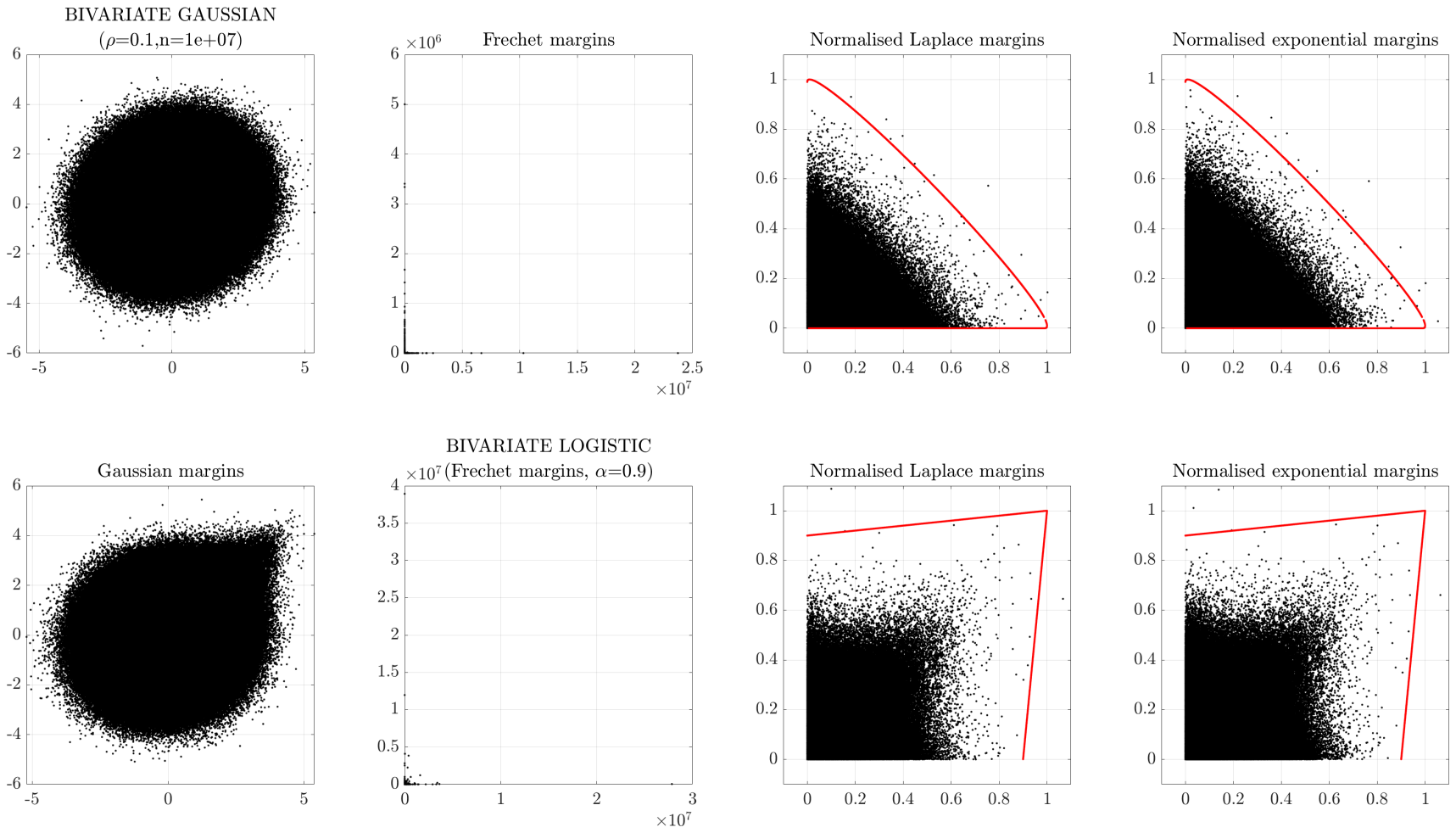
- Bivariate Gaussian: $f(x,y) = \phi(x,y;\mathbf{0},[1,\rho;\rho,1])$, where ϕ is the bivariate Gaussian density with correlation $\rho \in [-1,1]$
- Bivariate logistic: $C(u,v) = \exp\left(-\left[(-\log u)^{1/\alpha} + (-\log v)^{1/\alpha}\right]^{\alpha}\right)$, where dependence $\alpha \in (0,1]$

Extremal dependence



- Frechet $F(x) = \exp(-1/x)$, $x > 0$; Laplace $F(x) = \exp(x)/2$, $x < 0$ and $= 1 - \exp(-x)/2$, $x \geq 0$; exponential $F(x) = 1 - \exp(-x)$, $x \geq 0$
- Gaussian never touches (1,1) in normalised space, but the logistic can!
- What does this mean?

Extremal dependence



- Gaussian never touches (1,1) in normalised space, but the logistic can!

In other words ...

We have to think carefully about which kind of joint distribution (or copula) we assume when modelling multivariate extremes. Some copulas allow variables to get big together (e.g. bivariate logistic). Other copulas do not (e.g. bivariate Gaussian). Wrong kind of copula $\Rightarrow$ possible bias

Copulas which allow variables to get big together exhibit ASYMPTOTIC DEPENDENCE (AD)

Copulas which DO NOT allow variables to get big together exhibit ASYMPTOTIC INDEPENDENCE (AI)

Conventional statistical approaches assume (unrealistic) componentwise maxima, and always lead to AD. But there are modern approaches which assume AI. The best approach would allow both AI and AD

The marginal scale on which we compare bivariate distributions can be really revealing about AD and AI

Next: how can I tell if my sample data exhibits AD or AI?

Relative growth of joint tail: linear scale

Bivariate logistic: Including leading decay terms in z for $\alpha \in (0,1)$:

$$\frac{P(Z_1 > z, Z_2 > z)}{P(Z_1 > z)} = (2 - 2^{\alpha}) + \frac{2^{\alpha}(2^{\alpha} - 1)}{2z} + \frac{2^{\alpha}(3 \cdot 2^{\alpha} - 2 \cdot 2^{2\alpha} - 1)}{12z^2} + \mathcal{O}\left(\frac{1}{z^3}\right)$$

which converges to $2 - 2^{\alpha}$ as $z \to \infty$. So the joint tail probability scales like the marginal tail probability: this is *asymptotic dependence*. Adopting an asymptotic dependence model assumes joint large values of Z_1 and Z_2.

Bivariate Gaussian: Including leading decay terms in x for $\rho \in (0,1)$:

$$\frac{P(X_1 > x, X_2 > x)}{P(X_1 > x)} = \frac{(1+\rho)^{3/2}}{x\sqrt{2\pi(1-\rho)}} \exp\left(-\frac{1-\rho}{2(1+\rho)}x^2\right)\left(1 - \frac{2\rho}{(1-\rho^2)x^2} + \mathcal{O}\left(\frac{1}{x^4}\right)\right)$$

which converges to 0 as $x \to \infty$. The joint tail decays more quickly than the marginal tail: this is *asymptotic independence*. Adopting an asymptotic independence model prevents joint large values of X_1 and X_2.

Relative growth of joint tail: log scale

Bivariate logistic: Including leading decay terms in z for $\alpha \in (0,1)$:

$$\frac{\ln P(Z_1 > z, Z_2 > z)}{\ln P(Z_1 > z)} = 1 - \frac{\ln(2-2^\alpha)}{\ln z} - \frac{2^\alpha(2^\alpha-1)}{2(2-2^\alpha)} \cdot \frac{1}{z \ln z} + \mathcal{O}\left(\frac{1}{z^2 \ln z}\right)$$

which converges to 1 as $z \to \infty$. Under asymptotic dependence, this ratio is not interesting since it does not reveal anything about the distribution.

Bivariate Gaussian: Including leading decay terms in x for $\rho \in (0,1)$:

$$\frac{\ln P(X_1 > x, X_2 > x)}{\ln P(X_1 > x)} = \frac{2}{1+\rho} + \left(\frac{4\rho}{1+\rho}\right)\frac{\ln x}{x^2} - \frac{K_\rho}{x^2} + \mathcal{O}\left(\frac{(\ln x)^2}{x^4}\right)$$

which converges to $2/(1+\rho)$ as $x \to \infty$, where the constant K_ρ is given by:

$$K_\rho = 2C_0 + \frac{2\ln(2\pi)}{1+\rho} = 2\ln\left(\frac{(1+\rho)^{3/2}}{2\pi\sqrt{1-\rho}}\right) + \frac{2\ln(2\pi)}{1+\rho}.$$

Now, under asymptotic independence, looking at the ratio of the log probabilities reveals further structure in the decay of the joint tail probability relative to the margin. This is the motivation for statistic $\bar{\chi}$, the Ledford-Tawn model (and the statistic η) and the conditional extremes model.

In other words ...

Looking at the ratio of joint to marginal tail probabilities tells us about asymptotic dependence. We use asymptotic dependence assumptions when we expect to see simultaneous very large values of variables

Looking at the ratio of LOGS of joint to marginal tail probabilities tells us about asymptotic independence. We use asymptotic independence assumptions when we DON'T expect to see simultaneous very large values of variables

It's important to appreciate this key difference, otherwise we might get things wrong big time

The log-scale ratio underpins lots of statistical tools for asymptotic independence: $\bar{\chi}$, the Ledford-Tawn model (and the statistic η) and the conditional extremes model (used in covXtreme)

Next: how can I tell if my sample data exhibits AD or AI?

Asymptotic dependence (AD) and asymptotic independence (AI)

- Intuition
  - AD: both variables tend to get large together
  - AI: variables tend not to get large together

- Bivariate Gaussian exhibits AI for all correlations $\rho < 1$
- Bivariate logistic exhibits AD for all dependences $\alpha < 1$

- When we fit multivariate extreme value models, we must be able to distinguish AD and AI
  - Lots of copula models assume AD, and may be inappropriate for many environmental extreme value applications
  - Using an inappropriate dependence model causes bias
  - Classical multivariate extreme value theory (using "componentwise maxima") always leads to AD
- "Mea copula"
  - Financial crash 2007-8 (see MacKenzie and Spears 2014)
  - Proliferation of Gaussian copulas (assuming AI inappropriately)

Quantifying AD and AI

Variables $U = F_X(X)$ and $V = F_Y(Y)$ on uniform margins

χ quantifies AD

- $\chi(u) = \mathbb{P}(V > u \mid U > u) = \frac{\mathbb{P}(U>u, V>u)}{\mathbb{P}(U>u)} = 2 - \frac{\log C(u,u)}{\log(u)}$
- $\chi(u) \to \chi$ as $u \to \infty$
- $\chi = 1$: perfect dependence
- $\chi \in (0,1)$: AD
- $\chi = 0$: AI or perfect independence

$\bar{\chi}$ quantifies AI

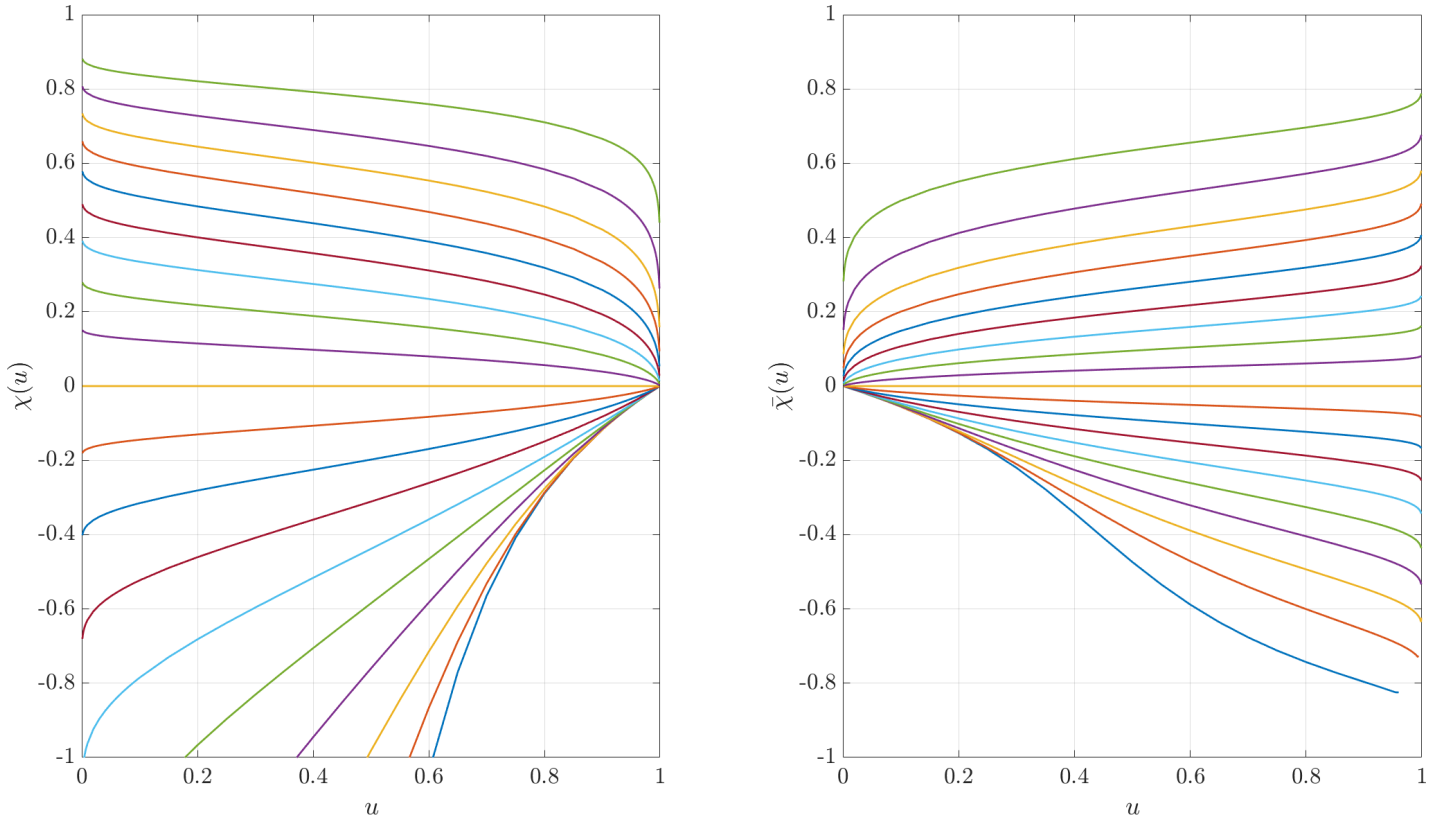
- $\bar{\chi}(u) = 2\frac{\log \mathbb{P}(U>u)}{\log \mathbb{P}(U>u, V>u)} - 1$
- $\bar{\chi}(u) \to \bar{\chi}$ as $u \to \infty$
- $\bar{\chi} = 1$: AD or perfect dependence
- $\bar{\chi} \in (0,1)$: AI
- $\bar{\chi} = 0$: perfect independence

Practical implication

- Estimate $\chi(u)$ and $\bar{\chi}(u)$ from data to test for AD and AI

AI in practice

- Many environmental extremes problems involve AI, at least in part ... bivariate Gaussian is the obvious example
- But convergence of $\chi(u)$ and $\bar{\chi}(u)$ to χ and $\bar{\chi}$ is very slow!



$\chi(u)$ and $\bar{\chi}(u)$ for bivariate Gaussian ($\Rightarrow \chi = 0, \bar{\chi} = \rho$); colours are correlations ρ on -0.9, -0.8, ..., 0.9 (Recreated from Coles et al. 1999)

In other words ...

$\chi(u)$ and $\bar{\chi}(u)$ provide simple ways of estimating whether a sample exhibits AD or AI. $\chi(u)$ compares the probability of two variables being big, with the probability of one variable being big, and is useful for AD. $\bar{\chi}(u)$ compares the LOGARITHMS of the same probabilities, and is useful for AI.

In practice, $\chi(u)$ and $\bar{\chi}(u)$ are very slow to converge to their asymptotic values for some distributions, like the bivariate Gaussian. So we have to be careful what we're doing

Next: a simple model based on regression that takes a lot of the worry away!

Conditional extremes

- Conditional extremes gives a straightforward approach to estimating multivariate tails, admitting both AD and AI
- Used widely in flood modelling

- $\boldsymbol{X} = (X_1, ..., X_j, ..., X_p)$ consists of p random variables
- Each X and Y have standard Laplace margins
  - $f(x) = \exp(-|x|)/2, x \in \mathbb{R}$
- Seek a model for $\boldsymbol{X}|(Y = y)$ for $y > u$

- Assume we can find p-dimensional scaling $\boldsymbol{a}, \boldsymbol{b} > \boldsymbol{0}$ such that

$$\mathbb{P}(\boldsymbol{Z} \leq \boldsymbol{z}|Y = y) \quad \to \quad G(\boldsymbol{z}) \quad \text{as} \quad u \to \infty$$
$$\text{for} \quad \boldsymbol{Z} \quad = \quad \frac{\boldsymbol{X} - \boldsymbol{a}(y)}{\boldsymbol{b}(y)}$$

- Non-degenerate G is unknown, and estimated empirically
- Similar thinking to that used to stabilise distribution of maxima

Conditional extremes

- Typical scaling is $\boldsymbol{a} = \boldsymbol{\alpha} y$ and $\boldsymbol{b} = y^{\boldsymbol{\beta}}$, $\boldsymbol{\alpha} \in [-1, 1]^p$, $\boldsymbol{\beta} \in (-\infty, 1]^p$
- So simply fit regression model

$$\boldsymbol{X} | (Y = y) = \boldsymbol{\alpha} y + y^{\boldsymbol{\beta}} \boldsymbol{Z} \quad \text{for} \quad y > u$$

- $\alpha = 1, \beta = 0$: AD or perfect dependence
- $\alpha \in (0, 1)$: AI

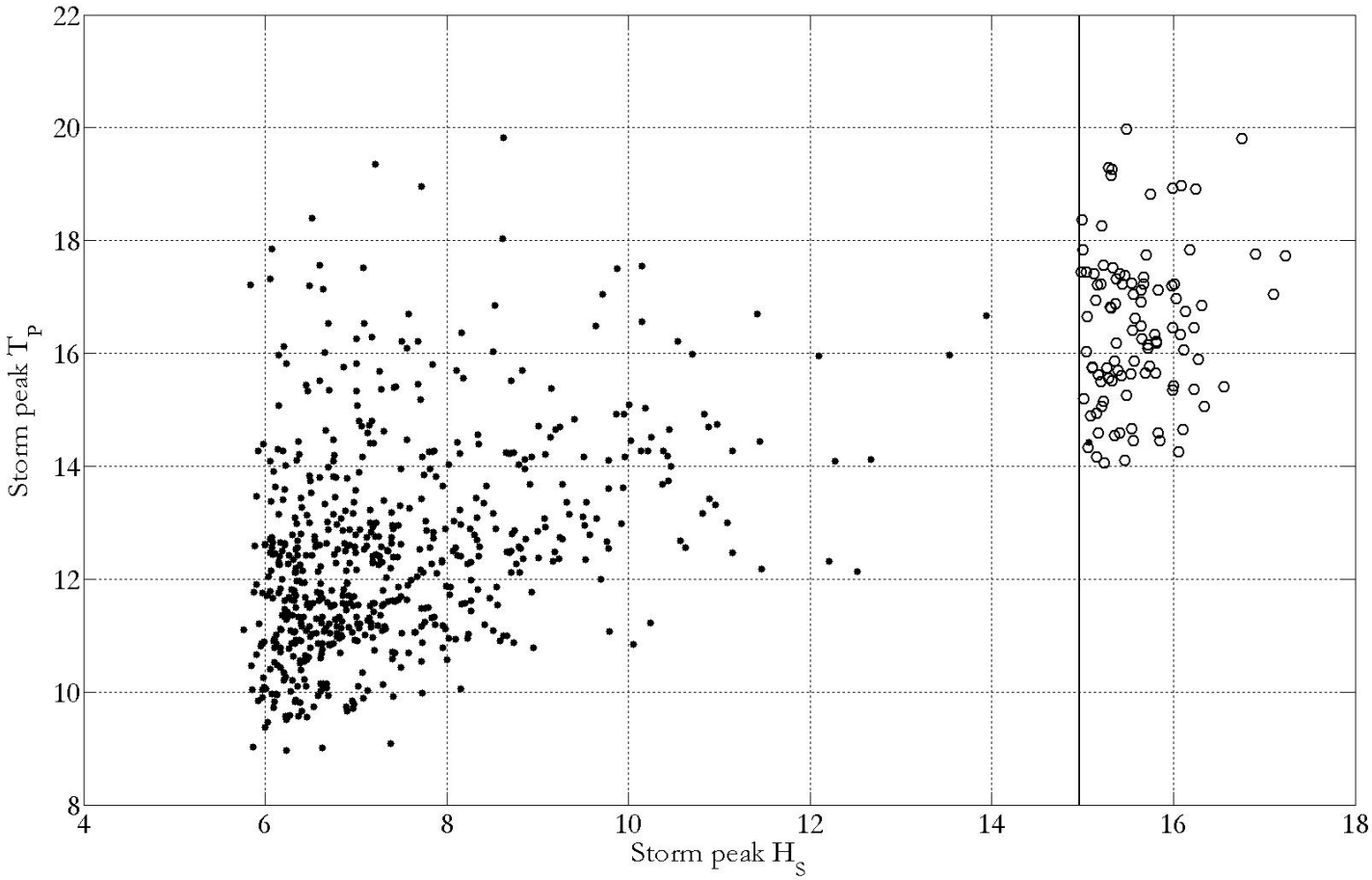
- Heffernan and Tawn [2004] find choices for α and β for popular bivariate cases; e.g. bivariate Gaussian : $\alpha = \rho^2, \beta = 1/2$
- Simulate under the fitted model to predict e.g.
  - Conditional distribution of T_P^{sp} given than H_S^{sp} exceeds its 1,000-year value
  - Joint conditional distribution of H_S^{sp} and T_P^{sp} given than U_{10}^{sp} exceeds its 1,000-year value

Fitting a simple conditional extremes model

- Fit models to physical-scale $\dot{X}$ and $\dot{Y}$
  - We need full-sample marginal models; we could use gamma-GP
- Transform sample from physical-scale to Laplace-scale X and Y
  - Using $F_{\text{Laplace}}(x) = F_{\text{GmmGP}}(\dot{x})$ (called the probability integral transform)
- Fit the model $X|(Y = y) = \alpha y + y^{\beta}(\mu + \sigma W)$ for $y > u$
  - Explore stability of extimated α as a function of u
  - Since distribution of W is unknown, assume it is standard Gaussian for model fitting, then estimate it empirically from fit residuals for simulation purposes
- Simulate new values of Y and $X|(Y = y)$ using the fitted model on Laplace scale
- Transform the simulated data back to physical scale to estimate conditional return values etc.

Illustrative simulation

- Sample of H_S^{sp} and T_P^{ass}
- Want to estimate the distribution of T_P^{ass} for values of H_S^{sp} much larger than those seen in the data



Fitting the simple conditional extremes model

- W assumed to follow a standard Gaussian
- Density for a single pair x, y on Laplace scale, with $y > u$

$$f_{\text{CE}}(x|y, \alpha, \beta, \mu, \sigma) = \phi(x; \alpha y + \mu y^{\beta}, (\sigma y^{\beta})^2)$$

  with parameters $\alpha \in [-1, 1]$, $\beta \in (-\infty, 1]$, $\mu \in \mathbb{R}$ and $\sigma > 0$
- Sample likelihood for pairs $\{x_i, y_i\}_{i=1}^{n}$ with $y_i > u$

$$\mathcal{L}_{\text{CE}}(\alpha, \beta, \mu, \sigma|\boldsymbol{x}, \boldsymbol{y}) = \prod_{i=1}^{n} f_{\text{CE}}(x_i|y_i, \alpha, \beta, \mu, \sigma)$$

- Maximum likelihood estimation

In other words ...

The conditional extremes model works for AD ($\alpha = 1$, $\beta = 0$) and AI ($\alpha < 1$), so we don't need to worry about the "dependence class". To use it, we first need to transform our sample (independently per variable) to standard Laplace scale; we need our covariate modelling tools to do this.

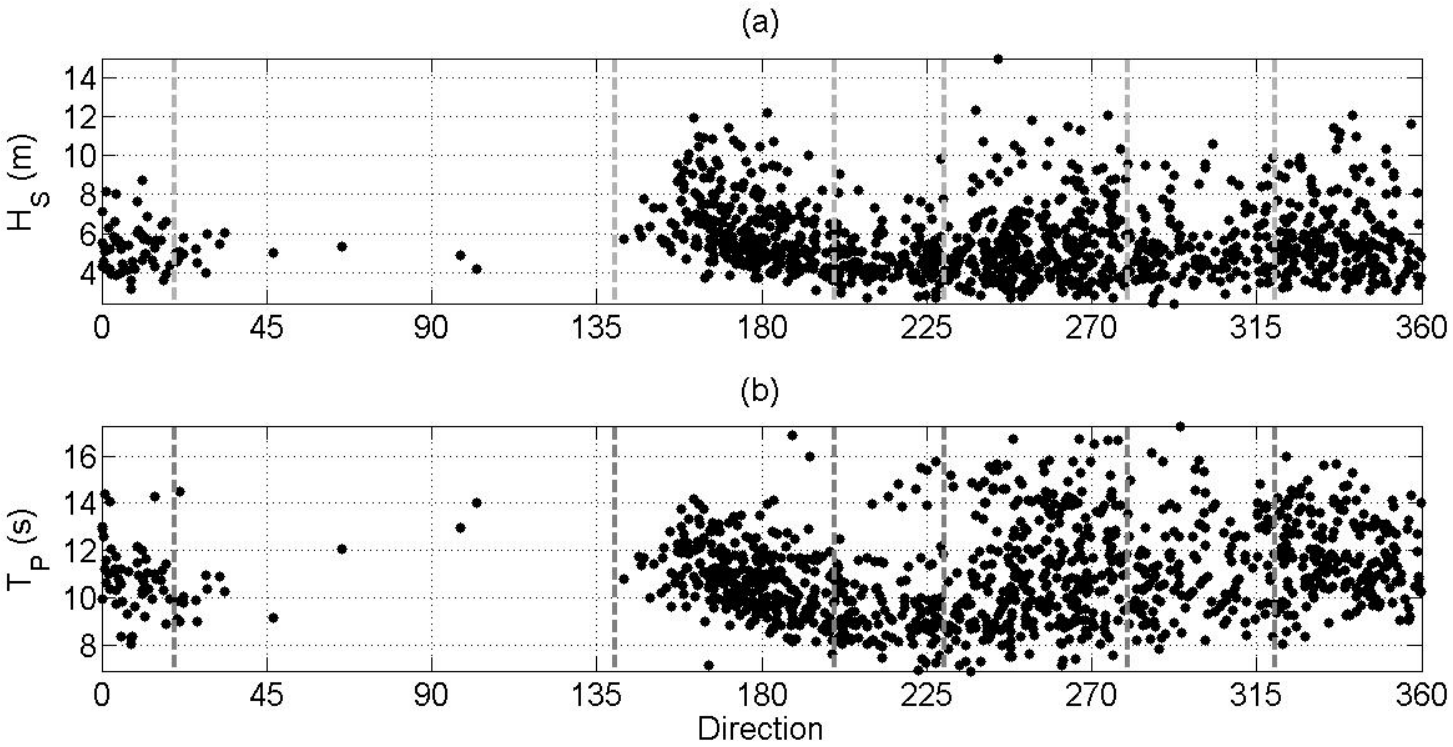
On Laplace scale, fitting the conditional extremes model is easy using maximum likelihood estimation. We assume that the residual distribution is Gaussian (or similar) for fitting, then use the actual residuals from the fit in simulation corresponding to very long return periods

We need to check that the estimated α is stable with threshold u. We also check that the residuals $x_i - \hat{x}_i$ are independent of the conditioning value y_i

Next: introducing covariates into the conditional extremes model

Non-stationary conditional extremes

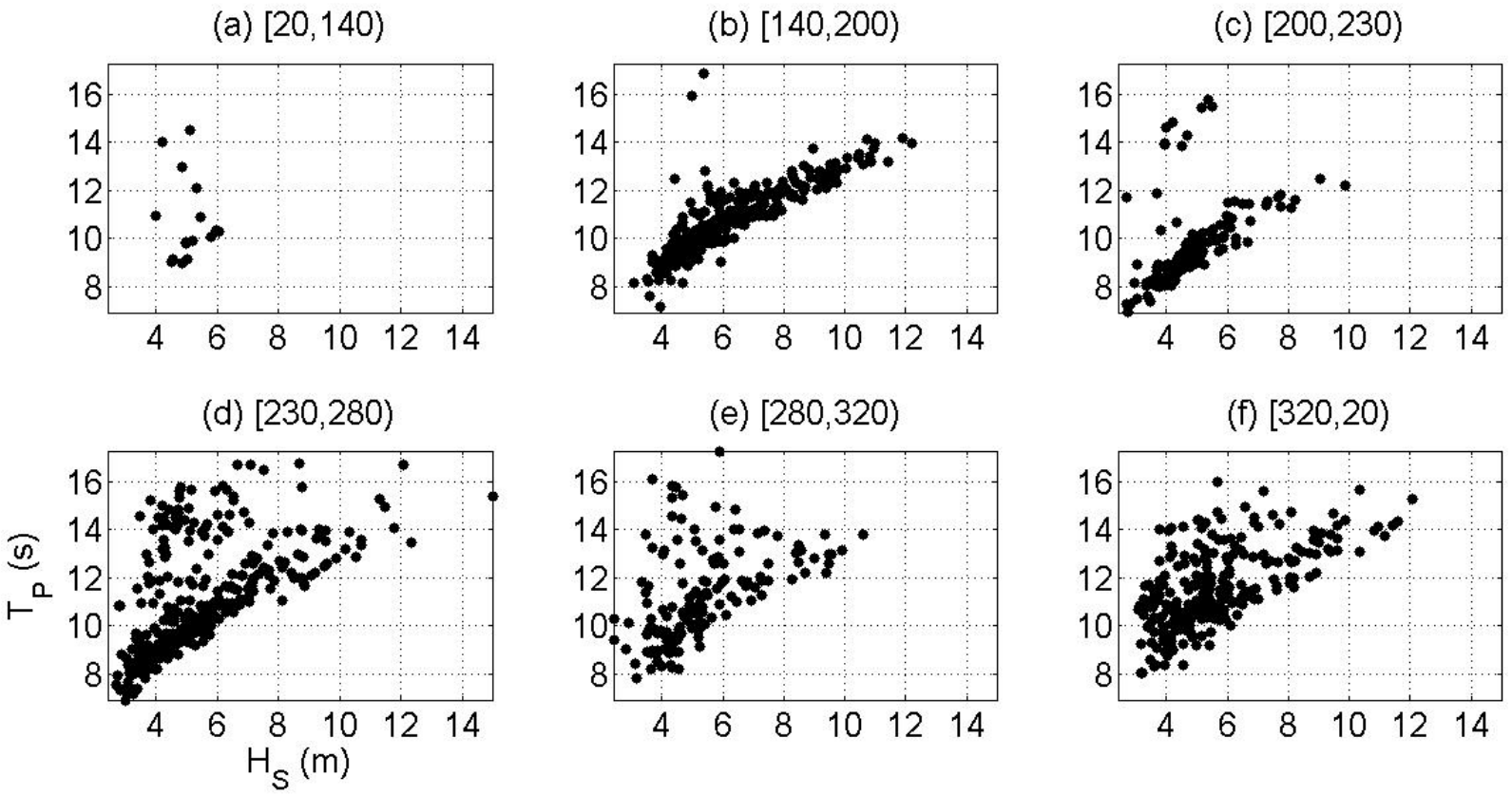
- We can allow α, β, μ and σ to vary with direction (say)
- Example from northern North Sea
- Details in Jonathan et al. [2014]



H_S^{sp} and T_P^{ass}

Non-stationary conditional extremes

- Dependence between H_S^{sp} and T_P^{ass} varies with direction



Fitting the non-stationary conditional extremes model

- α, β, μ and σ now vary with direction θ on $\mathcal{D}$
- Density for a single pair (x_i, y_i) with $y_i > u$

  $$f_{\text{CE}}(x_i|y_i, \alpha_{\mathcal{I}(i)}, \beta_{\mathcal{I}(i)}, \mu_{\mathcal{I}(i)}, \sigma_{\mathcal{I}(i)}) = \phi(x_i; \alpha_{\mathcal{I}(i)} y + \mu_{\mathcal{I}(i)} y^{\beta_{\mathcal{I}(i)}}, (\sigma_{\mathcal{I}(i)} y^{\beta_{\mathcal{I}(i)}})^2)$$

- Sample likelihood for pairs $\{x_i, y_i\}_{i=1}^n$ with $y_i > u$

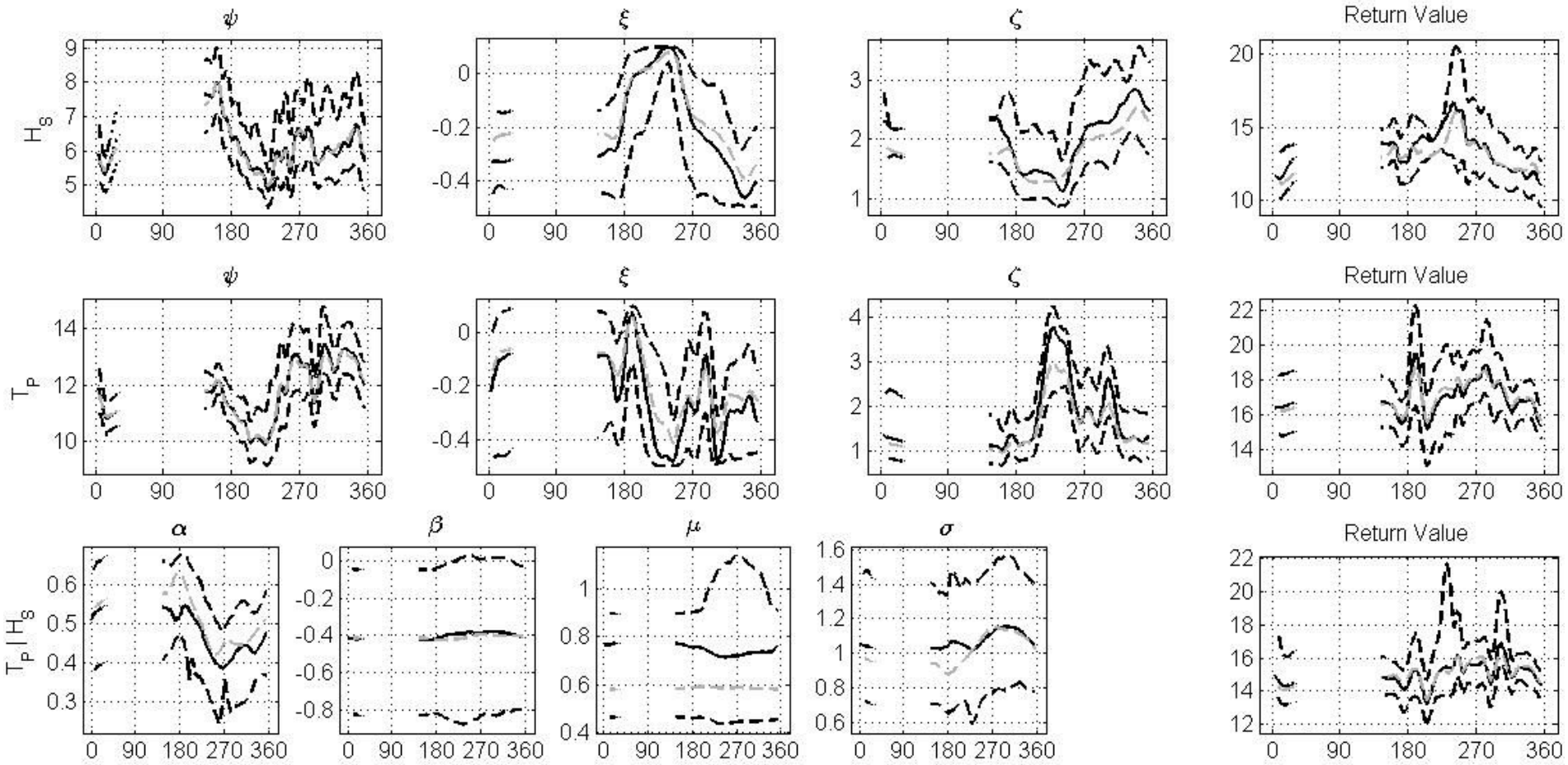
  $$\mathcal{L}_{\text{CE}}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\mu}, \boldsymbol{\sigma}|\boldsymbol{x}, \boldsymbol{y}) = \prod_{i=1}^{n} f_{\text{CE}}(x_i|y_i, \alpha_{\mathcal{I}(i)}, \beta_{\mathcal{I}(i)}, \mu_{\mathcal{I}(i)}, \sigma_{\mathcal{I}(i)})$$

  for parameter vectors $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$, with $\boldsymbol{\alpha} = \{\alpha_j\}_{j=1}^m$ etc.
- Penalised likelihood

  $$\mathcal{L}^*_{\text{CE}}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\mu}, \boldsymbol{\sigma}|\boldsymbol{x}, \boldsymbol{y}) = \mathcal{L}_{\text{CE}}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\mu}, \boldsymbol{\sigma}|\boldsymbol{x}, \boldsymbol{y}) \exp\left\{ -\sum_{\eta \in \{\alpha,\beta,\mu,\sigma\}} \lambda_\eta \boldsymbol{\gamma}_\eta' \boldsymbol{D}' \boldsymbol{D} \boldsymbol{\gamma}_\eta \right\}$$

  for spline coefficients $\boldsymbol{\gamma}$ and roughness coefficients λ
- Maximise $\mathcal{L}^*_{\text{CE}}$, using cross-validation to select λ_α, λ_β, λ_μ and λ_σ

Parameter estimates



Summary of marginal and CE parameters with direction for fitted model

- Marginal threshold ψ, GP shape ξ and GP scale ζ
- Note difference between marginal and conditional return values for T_P^{ass}

In other words ...

We can use the tools used previously for covariates with the conditional extremes model also: penalised maximum likelihood estimation, with cross-validation to estimate optimal spline roughness, and bootstrapping to quantify uncertainty

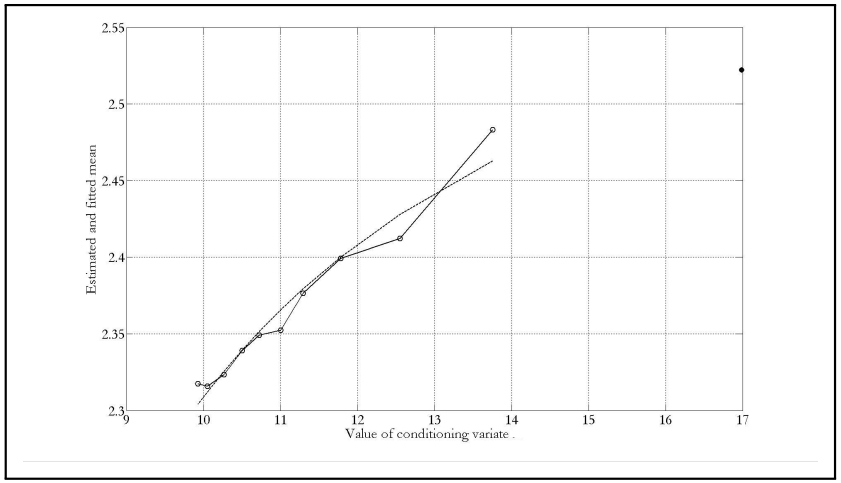
Estimating covariate variation in α is relatively straightforward: "the sample is relatively informative for α". However, spotting non-stationarity of β (and μ, σ) is more difficult. This suggests a simple conditional extremes model would allow α to vary with covariates, but nothing else $\Rightarrow$ covXtreme

Diagnostics for the non-stationary conditional extremes model are simply extensions of those for the stationary model

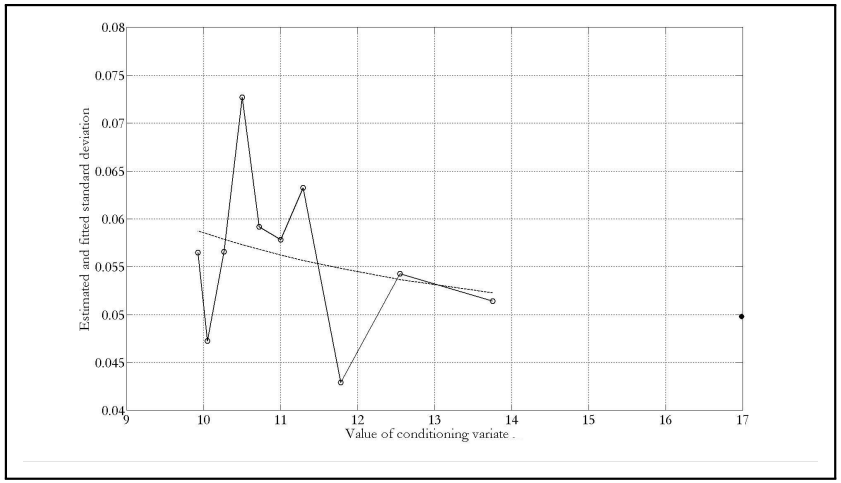
Weibull-log normal model

- Developed explicitly for joint modelling of $Y(H_S)$ and $X(T_P)$ for large values of Y
  - See e.g. Haver [1987]
- Assumes that $X|\{Y = y\}$ follows a log-normal distribution:
  - $f_{X|Y}(x|y) = \frac{1}{\sqrt{2\pi}\sigma(y)x} \exp(-\frac{(\log x - \mu(y))^2}{2\sigma^2(y)})$
  - Estimates for parameters $\mu(y)$ and $\sigma(y)$ then regressed on y to extrapolate
  - Some explicit forms for $\mu(y)$ recommended
- Assumes Weibull model for Y over a suitably large threshold
  - We consider a modified approach in which the marginal model for Y is assumed GP

Extrapolating estimates for μ and σ in Haver model



$\mu(y)$



$\sigma(y)$

- Justification for functional forms of $\mu(y)$ and $\sigma(y)$ is empirical
  - Extrapolation is problematic
  - Yields more biased and variable estimates of extreme quantiles; see Jonathan et al. [2010]
- The conditional extremes model has a better theoretical underpinning

In other words ...

The assumptions behind the Weibull log-normal model for H_S-T_P are harder to justify than those behind conditional extremes. Is H_S really Weibull? Is T_P given H_S really log-normal? The equations for extrapolating the log-normal mean and variance to long return periods are arbitrary

The conditional extremes model is more flexible, and given enough data, probably more likely to be "correct". But for small samples, both have problems ... all models are wrong ...

In ignorance, the best approach is to evaluate both approaches side by side using a cross-validation scheme for a given application, and use the one that gives best out-of-sample predictions. If you honestly don't know which is better, then use them both in a model average

Conditional extremes: what can go wrong?

- Marginal transformation required $\Rightarrow$ all the issues associated with marginal modelling and covariates
- Selection of appropriate threshold(s) u the dependence modelling
  - Use diagnostics e.g. $\hat{\alpha}(u)$ on u
  - Model averaging probably better when choice of u not obvious
- Small sample size is a particular problem
- Complex tail structures requiring multiple covariates or very high u
- Slow convergence to asymptotic form

Implications

- Propagate and quantify uncertainty carefully
  - No excuses!
- Model averaging in place of hyper-parameter selection
- Use cross-validation and bootstrapping in frequentist analysis; otherwise use Bayesian inference
- See Towe et al. [2023] for quantification of uncertainties in estimation of associated values

All models are wrong, some models are useful

If you want to read more

Metocean-related extensions of conditional extremes

- Non-stationary conditional extremes: Jonathan et al. [2014]
- Spatial conditional extremes: Shooter et al. [2019], Shooter et al. [2021b], Shooter et al. [2021a], Shooter et al. [2022]
- Time-series conditional extremes: Winter and Tawn [2016], Winter and Tawn [2017], Tendijck et al. [2019], Tendijck et al. [2024]

Classic multivariate extreme value analysis

- A gentle introduction: Coles [2001]
- Copulas: Joe [2014]
- Spatial extremes: Padoan et al. [2010]

Metocean applications with angular-radial models

- Murphy-Barltrop et al. [2025], Mackay et al. [2025]
- More in final session, hopefully

References

S. Coles. *An introduction to statistical modelling of extreme values*. Springer, Berlin, 2001.

S Coles, J Heffernan, and J Tawn. Dependence measures for extreme value analyses. *Extremes*, 2:339–365, 1999.

S. Haver. On the joint distribution of heights and periods of sea waves. *Ocean Eng.*, 14:359–376, 1987.

J. E. Heffernan and J. A. Tawn. A conditional approach for multivariate extreme values. *J. Roy. Statist. Soc. B*, 66:497–546, 2004.

H. Joe. *Dependence modelling with copulas*. CRC Press, 2014.

P. Jonathan, J. Flynn, and K. C. Ewans. Joint modelling of wave spectral parameters for extreme sea states. *Ocean Eng.*, 37: 1070–1080, 2010.

P. Jonathan, K. C. Ewans, and D. Randell. Non-stationary conditional extremes of northern North Sea storm characteristics. *Environmetrics*, 25:172–188, 2014.

E. Mackay, C. J. R. Murphy-Barltrop, J. Richards, and P. Jonathan. Deep learning joint extremes of metocean variables using the SPAR model. *J. Offshore Mech. Arct. Eng.*, 148:021201, 2025.

D. MacKenzie and T. Spears. A device for being able to book P&L: the organizational embedding of the Gaussian copula. *Soc. Stud. Sci.*, 44:418–440, 2014.

C. J. R. Murphy-Barltrop, E. Mackay, and P. Jonathan. Inference for bivariate extremes via a semi-parametric angular-radial model. *Extremes*, 28:209–238, 2025.

S. A. Padoan, M. Ribatet, and S. A. Sisson. Likelihood - based inference for max - stable processes. *J. Am. Statist. Soc.*, 105:263–277, 2010.

R. Shooter, E. Ross, J. A. Tawn, and P. Jonathan. On spatial conditional extremes for ocean storm severity. *Environmetrics*, 30:e2562, 2019.

R. Shooter, E Ross, A. Ribal, I. R. Young, and P. Jonathan. Spatial conditional extremes for significant wave height from satellite altimetry. *Environmetrics*, 32:e2674, 2021a.

R Shooter, J A Tawn, E Ross, and P Jonathan. Basin-wide spatial conditional extremes for severe ocean storms. *Extremes*, 24: 241–265, 2021b.

R. Shooter, E Ross, A. Ribal, I. R. Young, and P. Jonathan. Multivariate spatial conditional extremes for extreme ocean environments. *Ocean Eng.*, 247:110647, 2022.

S. Tendijck, E. Ross, D. Randell, and P. Jonathan. A non-stationary statistical model for the evolution of extreme storm events. *Environmetrics*, 30:e2541, 2019.

S. Tendijck, P. Jonathan, D. Randell, and J. A. Tawn. Temporal evolution of the extreme excursions of multivariate kth order Markov processes with application to oceanographic data. *Environmetrics*, 35:e2834, 2024.

R Towe, D Randell, J Kensler, G Feld, and P Jonathan. Estimation of associated values from conditional extreme value models. *Ocean Eng.*, 272:113808, 2023.

H. C. Winter and J. A. Tawn. Modelling heatwaves in central France: a case-study in extremal dependence. *J. Roy. Statist. Soc. C*, 65:345–365, 2016.

H. C. Winter and J. A. Tawn. kth-order Markov extremal models for assessing heatwave risks. *Extremes*, 20:393–415, 2017.